

Network Detection & Response (NDR) Architecture & Technical Specification



1. Executive Summary & Design Rationale

Modern enterprise networks face sophisticated threat actors employing living-off-the-land techniques, polymorphic command-and-control (C2) channels, and zero-day protocol tunneling. Traditional signature-only Intrusion Detection Systems (e.g., legacy Snort/Suricata without behavioral layers) fail against evasive variants, while standalone machine learning models suffer from catastrophic false-positive fatigue.

This platform implements a **fail-secure, dual-path architecture**:

- **Fast Signature Path**: Suricata rule engine executing against Layer 4-7 protocol payloads for known CVEs and high-rate anomalies.
- **Analytical Behavioral Path**: Supervised XGBoost multi-class classifier + Unsupervised PyTorch Autoencoder calibrated strictly on benign baseline network telemetry.

- **Fail-Secure Arbiter:** A hard defensive guarantee that any component timeout, memory fault, or unhandled exception defaults to immediate containment (`BLOCK_AND_ISOLATE` with `fail_secure=True`)—**never a silent pass.**

2. End-to-End Pipeline Architecture

```
flowchart TD
    subgraph Ingestion ["1. Traffic Ingestion Layer"]
        A["Raw Traffic / Replayed PCAPs"] --> B["Zeek Sensor"]
        A --> C["Suricata Sensor"]
        B --> D["ZeekParser (conn/dns/ssl/http)"]
        C --> E["SuricataParser (eve.json)"]
    end

    subgraph FeatureEng ["2. Feature Engineering"]
        D --> F["FlowFeatureExtractor (31+ Tabular Features)"]
        D --> G["Shannon & DNS Subdomain Entropy"]
        D --> H["Port Distribution Entropy"]
        D --> I["Inter-Arrival Timing Stats"]
        F & G & H & I --> J["NormalizedFlow Common Schema"]
    end

    subgraph DualPath ["3. Dual-Path Detection Engine"]
        E --> K["Suricata Signature Engine (ET Rules)"]
        J --> L["Supervised Flow Classifier (XGBoost)"]
        J --> M["Benign Baseline Autoencoder (PyTorch MSE)"]
        K --> N["Signature Matches | Detection Signals"]
        L --> N
        M --> N
    end

    subgraph Decision ["4. Fail-Secure Decision Engine"]
        N --> O["Decision Arbiter"]
        P["Timeout & Crash Watchdog"] -.-> O
        O --> Q["Verdict Selection"]
        Q --> R["Confidence >= 0.85 / Critical Sig"]
        Q --> S["Suspicious / Anomaly"]
        Q --> T["Low Risk / Benign"]
        Q --> U["Any Component Failure / Timeout"]
    end

    subgraph Response ["5. Containment & Visibility"]
        R & S --> V["ContainmentManager"]
        V --> W["Primary Action | OPNsense Firewall REST API"]
        V --> X["Failover Action | Local Host iptables Drop"]
        O --> Y["SIEM Dispatcher (ECS JSON)"]
        Y --> Z["Wazuh / ELK Dashboard"]
    end
```

3. Threat Detection Matrix & ATT&CK Alignment

MITRE ATT&CK ID	Technique Name	Primary Detection Path	Secondary Safety Net	Containment Verdict
T1046	Network Service Discovery	Suricata SID 3000001 (SYN Burst)	XGBoost PORT_SCAN Classifier	BLOCK_AND_ISOLATE
T1498.001	Direct Network Flood / DoS	Suricata SID 3000002 (SYN Flood)	XGBoost DDOS_FLOOD Classifier	BLOCK_AND_ISOLATE
T1071.001	Web Protocols C2 Beaconing	XGBoost C2_BEACONING	PyTorch Autoencoder MSE	QUARANTINE_VLAN
T1071.004	DNS Tunneling & Exfiltration	Shannon DNS Subdomain Entropy	Suricata SID 3000004	BLOCK_AND_ISOLATE
T1572	Protocol Tunneling / Evasion	PyTorch Benign Autoencoder (Zero-Day)	Suricata SID 3000005	BLOCK_AND_ISOLATE
T1110.001	Password Guessing / Brute Force	Suricata SID 3000006	XGBoost BRUTE_FORCE	BLOCK_AND_ISOLATE

4. Empirical Performance Guarantee

Evaluated against a **20% Stratified Holdout Split (20,002 test flows)** from the **CICIDS2017 dataset**:

- XGBoost Classifier**: Precision: **0.9998** | Recall: **0.9995** | F1: **0.9996** | FPR: **0.0004**
- PyTorch Benign Autoencoder**: Benign Specificity: **98.84%** (7,673 True Negatives / 90 False Positives out of 7,763 benign holdouts)
- Fail-Secure Latency**: Arbiter verdict resolved in **< 2.5ms** per flow.